

PART 1 : AI CONCEPTS :

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a technology that allows computers and machines to think and work like humans. AI can learn from data, solve problems, understand language, and make decisions. Today, AI is used in chatbots, voice assistants, self-driving cars, and many other applications.

2. Difference between AI, Machine Learning, Generative AI, and Agentic AI ?

- **Artificial Intelligence (AI):**
AI is the broad field of making machines intelligent so they can perform tasks that usually need human thinking.
- **Machine Learning (ML):**
Machine Learning is a part of AI where computers learn from data and improve without being directly programmed.
- **Generative AI:**
Generative AI creates new content such as text, images, music, videos, and code. ChatGPT is an example of Generative AI.
- **Agentic AI:**
Agentic AI can make decisions, plan actions, and complete tasks automatically to achieve a goal with less human help

3. Explain any three open-source AI models available on Hugging Face ?

- **Llama 3**
Llama 3 is an open-source language model developed by Meta. It can answer questions, write content, and help with coding tasks.
- **Mistral 7B**
Mistral 7B is a powerful and efficient AI model. It is known for giving good results while using fewer resources.

- **Stable Diffusion**

Stable Diffusion is an AI model that generates images from text descriptions. Users can create artwork and designs by simply typing prompts.

4. What is the difference between traditional programming and AI-assisted development ?

In traditional programming, developers write all the instructions that the computer follows. The computer only does what is programmed.

In AI-assisted development, AI tools help developers by suggesting code, finding errors, explaining concepts, and improving productivity. This makes software development faster and easier.

5. Explain AI hallucinations with one example ?

AI hallucination happens when an AI system gives incorrect or made-up information that sounds believable.

Example:

If a student asks an AI for a book written by a famous author, the AI may sometimes create a fake book title that does not actually exist. This is called an AI hallucination

